

Making and Qualifying Predictions from Scatter Plots

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|---|---|
| 1. 10 | predicted = 1 |
| 2. 70 | 8. 7.75 hours |
| 3. 75 | 9. (a) lowest spelling score = 60, — |
| 4. 6.75 hours | 10. 80 min |
| 5. (a) what the line predicts = 168, — | 11. (a) what the line predicts = -0.5 hours, — |
| 6. (a) sleep at 6 hours of TV = 6.5 hours, — | 12. (a) predicted score at 25 minutes = 63, (b) |
| 7. (a) predicted score = 84, (b) actual minus | difference in the two observed scores = 5, — |
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What is verified

14 of 19 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 5, 6, 9, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.SP.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 19 tagged checks.

- 1.** Display A: how many students are plotted?

Solution: Count the plotted points, not the tick marks: they sit at study times of 10, 20, 30, 35, 40, 50, 55, 60, 70 and 80 minutes. Ten points means ten students, and that is the whole evidence base for every prediction made from this display. 10

- 2.** Display A: the actual score of the student who studied 35 minutes.

Solution: Find 35 on the horizontal axis — it is the minor gridline halfway between 30 and 40 — go up to the point, and read across to the vertical axis. 70
Note for later: the trend line at $x = 35$ gives $0.6(35) + 48 = 69$, so this student sits one mark above the line.

- 3.** Display A: predict the score at 45 minutes.

Solution: Substitute $x = 45$ into $y = 0.6x + 48$: $0.6 \times 45 = 27$, and $27 + 48 = 75$. Because 45 lies inside the range of study times that were actually measured (10 to 80 minutes), this is an interpolation and is the kind of prediction the display supports. 75

4. Display B: predict sleep at 5.5 hours of television.

Solution: Substitute $x = 5.5$ into $y = 9.5 - 0.5x$: $0.5 \times 5.5 = 2.75$, and $9.5 - 2.75 = 6.75$. Again this is inside the measured range of 0 to 7 hours. 6.75 hours

5. Display A: (a) what the line gives for 200 minutes; (b) two reasons not to trust it.

Solution (a): $0.6 \times 200 = 120$, and $120 + 48 = 168$. 168

Solution (b) — instructor-judged. First, 200 minutes is far outside the range of study times anybody in the sample actually did, which stops at 80 minutes; the line has no evidence behind it out there, and nothing says the pattern continues. Second, the number itself is impossible — a quiz marked out of 100 cannot return 168, which is the display telling you the straight-line model has to bend somewhere. Full credit names the out-of-range extrapolation; naming only the impossible score earns half credit.

6. Display B: (a) sleep at 6 hours of television; (b) direction of the association.

Solution (a): Read up from 6 on the horizontal axis to the point, then across. 6.5 hours

Solution (b) — instructor-judged. The association is negative: as television time increases, sleep time decreases, and the points fall from upper left to lower right. For prediction that means any student with more television time should be given a *smaller* predicted sleep time — a prediction that went up with television time would contradict the display. Full credit names the direction and says what it implies for a prediction.

7. Display A: (a) predicted score at 60 minutes; (b) actual minus predicted.

Solution (a): $0.6 \times 60 = 36$, and $36 + 48 = 84$. 84

Solution (b): The student actually scored 85, so the gap is $85 - 84 = 1$ mark. A point one mark from the line is a close fit; the size of these gaps across the whole display is what tells you how much to trust a prediction. 1

8. Display B: mean sleep time of the eight students.

Solution: Add the eight sleep times and divide by eight: $9 + 9.5 + 8 + 8.5 + 8 + 6.5 + 6.5 + 6 = 62$, and $62 \div 8 = 7.75$. 7.75 hours

A prediction near the middle of the television range should land close to this value; one that does not is worth a second look.

9. Display C: (a) lowest spelling score; (b) can shoe size predict it?

Solution (a): The lowest plotted point sits at shoe size 5. 60

Solution (b) — instructor-judged. No. The points show no pattern at all: at shoe size 5 one student scored 60 and another 90, and the same wide spread appears at sizes 6, 7, 8 and 9. Knowing a student's shoe size narrows the possible spelling score hardly at all, so no trend line through these points would carry any predictive weight. Full credit refers to the absence of a trend in the plotted points.

10. Display A: study time the model calls for to reach a score of 96.

Solution: This runs the model backwards: solve $0.6x + 48 = 96$. Subtract 48 from both sides to get $0.6x = 48$, then divide both sides by 0.6: $x = 48 \div 0.6 = 80$. Eighty minutes is the largest study time anyone in the sample actually did, so this sits right at the edge of the evidence — reliable as a rough target, and the last point at which the model can be used at all without extrapolating.

80 min

11. Display B: (a) what the line predicts at 20 hours of television; (b) why it is not a prediction.

Solution (a): $0.5 \times 20 = 10$, and $9.5 - 10 = -0.5$.

-0.5 hours

Solution (b) — instructor-judged. A negative sleep time is impossible, which is the giveaway. The cause is extrapolation: television times of 20 hours are nowhere near the measured range of 0 to 7 hours, so the straight line is being read in territory where no data was collected and where the real relationship must flatten out. Full credit names the extrapolation beyond the data as the reason the number cannot be trusted.

12. Display A and the headline “Studying causes higher quiz scores”.

Solution (a): $0.6 \times 25 = 15$, and $15 + 48 = 63$.

63

Solution (b): $70 - 65 = 5$ marks.

5

Solution (c) — instructor-judged. No. The display shows an association — in this sample of ten students, longer study times came with higher scores — and an association cannot by itself establish a cause. Another explanation fits the same ten points just as well: for instance, students who already understand the material may find studying easier and so keep at it longer, or a generally motivated student may both study more and score higher, with motivation driving both. Ten students is also a small sample from one class. Full credit offers a specific alternative explanation; naming the association-is-not-causation rule with no alternative earns half credit.